# Yousef Alaa Awad

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## Education

**University of Central Florida** — Orlando, FL
Bachelor of Science in Computer Engineering with University Honors | GPA: 3.77 — Expected Aug 2027
Relevant Courses: Verification of Digital Systems, HDL in Digital Systems, and Adv. Computer Architecture
Affiliations: IEEE-HKN (Zeta Chi Chapter), IEEE, ACM, Burnett Honors College

## Work Experience

**Advanced Micro Devices, Inc.** — Austin, TX
Validation/Debug Intern — Sept 2026 – Present

- Performing system validation on client SoCs, triaging post-silicon failures across 2 validation teams; built a Python tool integrating Jira REST and Anthropic Claude APIs that handles the documentation start and Jira ticket creation for failure triage, cutting per-ticket triage time by 5–10 minutes.
- Provisioning 15–20 client validation platforms weekly by running multi-stage automated flashing flows with locked BIOS, OS, and driver baselines, eliminating configuration drift and isolating true hardware bugs from test-environment issues.

**Lockheed Martin Corporation** — Orlando, FL
ASIC/FPGA Design CWEP Intern — Feb 2026 – July 2026

- Built a Synopsys VC Formal + Tcl FPV flow (20+ SVAs) achieving full unbounded proofs on internal state and 500-cycle bounds on high-fanin outputs, using built-in vacuity checks and Synopsys VCS counterexample replay to debug 3 FSM bugs.
- Extended the formal framework with Formal Security Verification (FSV/SPV) by defining taint-propagation and non-interference properties for a cryptographic module, formally proving the datapath against unauthorized secret leakage.
- Deployed a modular, config-driven FPV/FSV regression suite in GitLab CI/CD using parameterized testbench templates and Tcl build scripts, reducing per-module formal bring-up time from ∼60 minutes to ∼5 minutes via single variable toggle.

Embedded Software Engineering CWEP Intern — Apr 2025 – Feb 2026

- Designed a system-level Hardware-in-the-Loop (HIL) simulation environment in C++, enabling high-fidelity testing within the existing workflow for 20+ software engineers by implementing a CIGI v2 compliant image generator.
- Remediated 250+ security violations flagged by static code analysis, resolving critical memory-safety issues including buffer overflows and unsafe pointer access across the codebase.
- Engineered the migration of legacy C++/Qt tooling environments from Visual Studio to a CMake build system, slashing compilation times by 50% and establishing a modular architecture to improve future scalability of the application.

## Projects

**KnightCore – RISC-V SoC – 2025 AMD Hardware Competition** – Group Project — Apr 2025 – Aug 2025

- Built a coverage-driven Python (Cocotb) testbench for a 32-bit RISC-V SoC (RV32I), pairing a golden reference model and self-checking scoreboard to verify RTL state across constrained-random instruction streams.
- Developed a directed/constrained-random test suite targeting arithmetic corner cases, data hazards, and branch prediction logic across all RV32I instructions, closing functional testplan targets and isolating 5 critical RTL bugs prior to synthesis.

**2025 SouthEastCon Robotics Competition – Embedded Software Lead** – Group Project — Sep 2024 – Apr 2025

- Managed Git/GitHub workflows with branching strategies, code reviews, and issue tracking to ensure team-wide consistency. Built CI/CD pipelines for automated testing and deployment, and implemented HIL testing for early hardware validation.
- Led the embedded software pipeline using ROS2, enabling low-latency control between modular hardware and decision nodes, which was then validated via Gazebo/OpenCV, contributing to 1st place in Design and 2nd place in Performance.

**CyndaQuil Compiler/Language** – Solo Project — Sep 2024 – Jan 2025

- Designed and implemented a statically-typed compiled language from scratch in C++, building the full toolchain (parser, semantic analyzer, and x86 codegen) with CMake managing a multi-stage build pipeline.
- Engineered low-level system components and optimized performance, integrating Assembly and C to enhance hardware-software interfacing, ensuring minimal latency and efficient computation for embedded systems.

## Leadership

**Eta Kappa Nu (IEEE-HKN)**
Chapter President, Zeta Chi Chapter — Apr 2026 – Present

- Directing operations and budget for a 40+ member chapter following Key Chapter recognition, expanding induction classes, driving alumni re-engagement, and scaling member programming for long-term sustainability.

## Technical Skills

**Verification & Formal Methods**: Synopsys VC Formal, SystemVerilog Assertions (SVA), UVM (Coursework), Constrained-Random Verification, Functional Coverage, Cocotb, Bounded Model Checking, Formal Property Verification (FPV), Formal Security Verification (FSV/SPV)
**Languages**: SystemVerilog, Verilog, C++, Tcl, Python, C, Bash, Rust, Assembly (x86, RISC-V, MIPS)
**Computer Architecture**: RISC-V ISA, AXI/Memory-Mapped Interfaces, Pipelining, Caching, Memory Hierarchy
**Hardware & EDA Tools**: Synopsys VCS, Xilinx Vivado, Verilator
**Systems & Scripting**: Linux, Git, GitLab CI/CD, ROS/ROS2, Real-Time C++, CMake/Make